

Instructor: Dr. Rajesh Kumble Nayak

10 Sep, 2025 @ 10:00 AM.

Duration 60+30 Min.

- Read each question carefully.
- Answer all questions.
- Ensure all parts of the answer for a given question are in the same place.
- Write neatly and do all rough work on the answer sheet.
- Do not write on the question paper.
- Keep answers concise and to the point.

Take a deep breath and relax.

Good luck!**Problem - 1 {15 Marks}:**In a generalised coordinate system $\{\xi, \eta\}$, the Lagrangian is given by

$$\mathcal{L} = \frac{m}{2} (\xi^2 + \eta^2) (\dot{\xi}^2 + \dot{\eta}^2) + \frac{2k}{\xi^2 + \eta^2}.$$

- Show that $C \equiv \xi p_\eta - \eta p_\xi$ a constant of motion. Here $\{p_\xi, p_\eta\}$ are generalized momenta corresponding to $\{\xi, \eta\}$.
- Is the total energy E conserved in this system? Can you identify the Kinetic energy and Potential Energy?
- Is the system integrable? If yes find the necessary constants of motion.

Problem 1generalized coordinate $\{\eta, \xi\}$

(a)

$$\mathcal{L} = \frac{m}{2} (\xi^2 + \eta^2) (\dot{\xi}^2 + \dot{\eta}^2) + \frac{2k}{\xi^2 + \eta^2}$$

$$\text{Let } C = \xi p_\eta - \eta p_\xi$$

$$p_{\dot{z}} = \frac{\partial L}{\partial \dot{z}} = m(\dot{z}^2 + \dot{\eta}^2) \dot{z}$$

$$p_{\dot{\eta}} = \frac{\partial L}{\partial \dot{\eta}} = m(\dot{z}^2 + \dot{\eta}^2) \dot{\eta}$$

Now

$$\frac{dC}{dt} = \dot{z} p_{\dot{\eta}} - \dot{\eta} p_{\dot{z}} + z \frac{dp_{\dot{\eta}}}{dt} - \eta \frac{dp_{\dot{z}}}{dt}$$

From Lagrange Eqn of motion we have

$$\frac{dC}{dt} = \dot{z} p_{\dot{\eta}} - \dot{\eta} p_{\dot{z}} + z \frac{\partial L}{\partial \eta} - \eta \frac{\partial L}{\partial z}$$

$$\frac{\partial L}{\partial \eta} = m(\dot{z}^2 + \dot{\eta}^2) \dot{\eta} - \frac{4k\eta}{(\dot{z}^2 + \dot{\eta}^2)^2} \quad (2)$$

$$\frac{\partial L}{\partial z} = m(\dot{z}^2 + \dot{\eta}^2) \dot{z} - \frac{4kz}{(\dot{z}^2 + \dot{\eta}^2)}$$

$$\begin{aligned}
 \frac{dC}{dt} &= m(\dot{z}^2 + \dot{\eta}^2) \dot{z} \dot{\eta} - m(\dot{z}^2 + \dot{\eta}^2) \dot{z} \dot{\eta} \\
 &+ m(\dot{z}^2 + \dot{\eta}^2) \dot{z} \dot{\eta} - \frac{4k\eta \dot{z}}{(\dot{z}^2 + \dot{\eta}^2)^2} \\
 &- m(\dot{z}^2 + \dot{\eta}^2) \dot{z} \dot{\eta} + \frac{4k\eta \dot{z}}{(\dot{z}^2 + \dot{\eta}^2)^2}
 \end{aligned}$$

$= 0$ hence C is constant

$$(b) \quad h = \sum p_i \dot{q}_i - L$$

$$= p_{\dot{z}} \dot{z} + p_{\dot{\eta}} \dot{\eta}$$

$$= p_{\dot{z}} \dot{z} + p_{\dot{\eta}} \dot{\eta} - \frac{m}{2} (\dot{z}^2 + \dot{\eta}^2) (\dot{z}^2 + \dot{\eta}^2)$$

$$- \frac{2k}{\dot{z}^2 + \dot{\eta}^2}$$

(3)

$$h = \frac{m}{2} (\dot{z}^2 + \dot{\eta}^2) (\dot{z}^2 + \dot{\eta}^2) - \frac{2k}{\dot{z}^2 + \dot{\eta}^2}$$

$$h = \frac{m}{2} (\dot{z}^2 + \dot{\eta}^2) (\dot{\bar{z}}^2 + \dot{\bar{\eta}}^2) - \frac{2k}{\dot{z}^2 + \dot{\eta}^2}$$

Since $\frac{\partial L}{\partial t} = 0 \Rightarrow h$ is const of motion.

We may write $L = T - V$

$$\text{where } T = \frac{m}{2} (\dot{z}^2 + \dot{\eta}^2) (\dot{\bar{z}}^2 + \dot{\bar{\eta}}^2)$$

$$V = - \frac{2k}{\dot{z}^2 + \dot{\eta}^2}$$

$$T = T_2 \quad \text{and} \quad V = -L_0$$

$$\text{hence } h = T + V = E$$

④
 (c) System is integrable because there are two independent-constant of motion E and C

Problem - 2 {15 Marks}:

A particle of mass m moves in the plane under the influence of a potential

$$V(r) = -\frac{k}{r} + \frac{\beta}{2r^2}, \text{ and } k > 0, \beta \in \mathbb{R}.$$

Here the coordinate system used is plane-polar $\{r, \phi\}$.

- Find two constants of motion.
- Find the equation of motion.
- Find the orbital equation of motion. (Yea, that one with $\frac{dr}{d\phi} = \dots$. oh! you may have to change the variable to $u = \frac{1}{r}$. Oh! one more point, to get full marks you might have get second order equation, something like $\frac{d^2 u}{d\phi^2} = \dots$)

coordinates $\rightarrow \{r, \phi\}$ (5)

Prob 2 : $V(r) = -\frac{k}{r} + \frac{\beta}{2r^2}$

$k > 0 \rightarrow \beta \in \mathbb{R}$

$$L_e = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\phi}^2) - V(r)$$

$$L_e = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\phi}^2) + \frac{k}{r} - \frac{\beta}{2r^2}$$

$\Rightarrow$ Constants of motion

Since $\frac{\partial L_e}{\partial \phi} = 0 \Rightarrow L_z \equiv \frac{\partial L_e}{\partial \dot{\phi}} = \text{const}$

$$\Rightarrow m r^2 \dot{\phi} = L_z$$

$$\dot{\phi} = \frac{L_z}{m r^2}$$

$$\dot{\phi} = \frac{L_z}{m r^2}$$

$$\frac{\partial L}{\partial t} = 0 \quad L_z = T_z \quad L_{\phi} = -V$$

$\Rightarrow$ Energy $E = T + V$ is constant of motion

$$E = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\phi}^2) - \frac{k}{r} + \frac{\beta}{2r^2}$$

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EOM is

$$\dot{\phi} = \frac{L_z}{m r^2}$$

the radial Eqn is

$$E = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\phi}^2) - \frac{k}{r} + \frac{\beta}{2r^2}$$

$$\begin{aligned} \frac{2}{m} \left(E + \frac{k}{r} - \frac{\beta}{2r^2} \right) &= \dot{r}^2 + r^2 \dot{\phi}^2 \\ &= \dot{r}^2 + \frac{L_z^2}{m^2 r^2} \end{aligned}$$

$$\dot{r}^2 = \frac{2}{m} \left[E + \frac{k}{r} - \frac{\beta}{2r^2} - \frac{L_z^2}{2m r^2} \right]$$

$$\frac{2}{m} \left(E + \frac{k}{r} - \frac{\beta}{2r^2} \right) = \dot{r}^2 + r^2 \dot{\phi}^2$$

$$= \dot{r}^2 + \frac{L_z^2}{m^2 r^2}$$

$$\dot{r}^2 = \frac{2}{m} \left[E + \frac{k}{r} - \frac{\beta}{2r^2} - \frac{L_z^2}{2mr^2} \right]$$

$$\dot{r} = \left[\frac{2}{m} \left(E + \frac{k}{r} - \frac{\beta}{2r^2} - \frac{L_z^2}{2mr^2} \right) \right]^{1/2}$$

The radial effective potential is

$$V_{\text{eff}} = -\frac{k}{r} + \frac{\beta}{2r^2} + \frac{L_z^2}{2mr^2}$$

find the orbit Equation!

(7)

$$\frac{d\phi}{dt} = \frac{L_z}{mr^2}$$

The radial Eqn is

$$\dot{r}^2 = \frac{2}{m} \left(E + \frac{k}{r} - \frac{\beta}{2r^2} - \frac{L_z^2}{2mr^2} \right)$$

$$\frac{dr}{dt} = \frac{dr}{d\phi} \cdot \frac{d\phi}{dt} = \frac{dr}{d\phi} \cdot \frac{L_z}{mr^2}$$

$$\left(\frac{dr}{d\phi}\right)^2 \frac{L_z^2}{m^2 r^4} = \frac{2}{m} \left[E + \frac{k}{r} - \frac{\beta}{2r^2} - \frac{L_z^2}{2mr^2} \right]$$

Now let $u = \frac{1}{r}$

$$\frac{du}{d\phi} = -\frac{1}{r^2} \frac{dr}{d\phi}$$

$$\left(\frac{du}{d\phi}\right)^2 r^4 \frac{L_z^2}{m^2 r^4} = \frac{2}{m} \left[E + \frac{k}{r} - \frac{\beta}{2r^2} - \frac{L_z^2}{2mr^2} \right]$$

$$\left(\frac{du}{d\phi}\right)^2 = \frac{2m}{L_z^2} \left[E + ku - \frac{\beta u^2}{2} - \frac{L_z^2}{2m} u^2 \right]$$

differentiate w.r.t ϕ again

(8)

$$\cancel{2} \frac{\cancel{du}}{\cancel{d\phi}} \cdot \frac{d^2 u}{d\phi^2} = \cancel{\frac{2m}{L_z^2}} \left[k - \frac{\beta u}{2} - \frac{L_z^2 u}{m} \right] \frac{\cancel{du}}{\cancel{d\phi}}$$

$$\frac{d^2 u}{d\phi^2} = \frac{km}{L_z^2} - \left(\frac{m\beta}{L_z^2} + 1 \right) u$$

(8)

differentiate w.r.t ϕ again

$$2 \frac{du}{d\phi} \cdot \frac{d^2 u}{d\phi^2} = \frac{2m}{L_z^2} \left[k - \frac{\alpha \beta u}{a} - \frac{L_z^2 u}{m} \right] \frac{du}{d\phi}$$

$$\frac{d^2 u}{d\phi^2} = \frac{km}{L_z^2} - \left(\frac{m\beta}{L_z^2} + 1 \right) u$$

$$\Rightarrow \frac{d^2 u}{d\phi^2} + \omega^2 u = \frac{mk}{L_z^2} \quad \text{is the orbit equation.}$$

$$\omega^2 = \frac{m\beta}{L_z^2} + 1$$